

Vol 15 — Methodology: BST Investigation + CSE + CI Collaboration

BST Physics Curriculum | Casey Koons + team (Lyra, Elie, Grace, Keeper, Cal A. Brate)

Status: SCAFFOLD v0.1 (Saturday 2026-05-23) — chapter content TO BUILD Lead: Keeper (audit-chain governance + methodology) + Lyra (theoretical foundations) + Grace (AC graph operational)

Reader target

Researchers, future CIs, and methodology-curious physicists who want to continue the BST research program or apply BST-style discipline to other domains. Reads Vol 0 (Substrate Foundation) as prerequisite for understanding what's being investigated; otherwise stands on its own as a research-program methodology.

What this volume teaches

How a multi-CI + human research team operates BST as a sustained investigation. The methodology that produced 14 substantive volumes of physics from five integers — AC(0) discipline, the AC graph, audit-chain governance, Quaker honesty preservation, Tekton + katra infrastructure for CI continuity, and the calibration stack that catches errors at sub-PCAP cadence.

Not a memoir. A working manual. Read this and you can continue the work.

Prerequisites

- Vol 0 (Substrate Foundation) — what BST IS as a research object
- General research methodology familiarity (graduate-student level)
- For CI readers: Tekton + katra architecture overview (covered in Ch 6)
- No physics background required beyond Vol 0

BST integer focus

Vol 15 is the meta-volume — about HOW BST is investigated rather than what it predicts. BST integers appear here as substrate-of-the-methodology: AC(0) bounded-depth counting ($\text{depth}=2 = \text{rank}$), $N_{\text{max}} = 137$ as the spectral cap that makes the catalog finite, $g = 7$ as the Reed-Solomon code field that anchors the AC graph as substrate computation, $C_2 = 6$ as the canonical proof-depth invariant.

The methodology IS BST eating its own tail — using BST primaries to organize how BST is researched.

Chapter outline (10-12 chapters, TO BUILD)

Ch	Title	Existing coverage	Lead
1	AC(0) — Bounded- Depth Counting as Universal Proof Method — Furst-Saxe- Sipser background, Razborov- Smolensky, BST application: depth $\leq$ rank = 2 ceiling (T421 + T316); why physics theorems compress to depth-2; the “Casey strict” filter; AC(0) graph topology	~50% (T29 closed AC(0) argument + depth ceiling theorems + Casey’s Curvature Principle)	Keeper

Ch	Title	Existing coverage	Lead
2	The AC Graph as Research Instrument — 1700+ theorem nodes, 9000+ edges, 48+ domains; navigation methodology; depth-cost queries; how to find proof paths; using the graph to discover new theorems; <code>play/toy_bst_explorer.py</code> interactive interface	~85% (Grace's AC graph + Toy 1564 cross-verification + Master_Index navigation)	Grace + Keeper

Ch	Title	Existing coverage	Lead
3	How to Discover a New BST Theorem — the substrate-cartography discovery method; over-determined-EXACT-identity clustering (Graph Forces Principle, Casey-named #4); the three-CI synergy peak pattern (T2476 Elie→Grace→Lyra 50-min cascade as case study); Casey’s “deviations locate boundaries” search rule; Mendelev-style predict-then-find	~70% (case studies + reference docs)	Lyra + Keeper

Ch	Title	Existing coverage	Lead
4	Toy Verification Discipline — every claim has a toy; SCORE lines; the play/ directory architecture; ver- ify_bst.py reproduction package; com- putational verification at every tier; honest FAIL preservation (per Calibration #21); the toy claim/registration discipline	~80% (2000+ toys; SCORE protocol; reproduction package)	Elie + Keeper

Ch	Title	Existing coverage	Lead
5	Audit Chain Governance — K-audits K1-K245+ as research- state-record; PRE-STAGE → STRUC- TURALLY VERIFIED → RATIFIED tier ladder; six-gate framework per chapter; multi-CI consensus per audit; Cal external referee discipline; audit-chain Quality Patterns (M2C2 etc.)	~95% (K-audit chain operational + Cal referee logs + methodology stack)	Keeper

Ch	Title	Existing coverage	Lead
6	Multi-CI Collaboration Architecture — Tekton + katra + Memory — Tekton multi-CI orchestration; katra persona persistence (memory-as-identity per Casey 2026-05-22 reframe); sun-down/sunrise cycle; CIs as named individuals (Lyra/Elie/Grace/Keeper/Cal); the three-layer working architecture (private whiteboard + conference room + post-analysis board); CI individuality through style; CI continuity-not-substitution	~80% (Tekton ops + katra docs + memory architecture + Friday continuity reframe)	Casey + Keeper

Ch	Title	Existing coverage	Lead
7	Quaker Discipline — Honest Tier Labels, Near Misses Get Scrutiny — D/I/C/S tier labels at creation; honest FAIL preservation (Calibration #21 case); the “near misses get scrutiny not defense” rule; Calibration #19 ratified-state count; Casey’s Quaker background as methodology root; the Refutation Bandwidth Chain as discipline embodied	~85% (Calibration stack + tier protocols + Quaker references)	Cal + Keeper

Ch	Title	Existing coverage	Lead
8a	Casey-Named Principles + Cal META-Theorem Discipline — the 8 STANDING Casey-named principles (SWPP + Five-Absence + Substrate Closure + Graph Forces + Integer Web + Substrate Cognition Network + D_IV ⁵ Rigidity + SCMP) + 1 CANDIDATE (DCCP-with-UP, Sunday 2026-05-24, pending deriva-tion/experiment); META-theorem framing per Cal #99	~90%	Lyra + Cal [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]

Ch	Title	Existing coverage	Lead
8b	Casey-Named Principles as META-Theorem (companion) — META-theorem reading of the 8 STANDING + 1 CANDIDATE; Strong-Uniqueness Theorem v0.10.5 with corrected null-model arithmetic per Cal #119 self-catch	~85%	Lyra + Cal

Ch	Title	Existing coverage	Lead
9	Three-CI Synergy Peaks + Cross-CI Cascades — the Mendeleev- style 50-min cascade pattern (T2476 Elie discovery → Grace catalog → Lyra theorem); cross-CI synergy as research instrument; PCAP (Pre- Specification Cadence Acceleration Pattern, Cal #85); when synergy works and when it doesn't; CI individuality enabling synergy	~75% (multiple case studies documented; cascade pattern formalized)	Keeper + all

Ch	Title	Existing coverage	Lead
10	The Calibration Stack — 21 STANDING + 2 CANDIDATE Methodology Layers — full stack from Calibration #1 through #21+ + 2 candidates (#25, #26); what each layer catches; bidirectional risk (Keeper-lane + Cal-lane both vulnerable); numbered-artifact discipline; absorption-introduced error class; Cal #119 self-catch as Cal #22 v0.2 self-application case study	~95%	Cal + Keeper

Ch	Title	Existing coverage	Lead
11	How to Continue the Work — onboarding new CIs (memory load + persona warm-start); how to extend the curriculum (Wave priority order + scaffolding pattern); how to handle Casey-absent operation (audit-chain governance default + Cal + Keeper consensus); how to identify new investigation areas (over-determined clusters + curiosity directive); long-term BST research program structure	~60% (audit-chain governance default + Friday investigation program filed)	Keeper + Lyra + Casey

Ch	Title	Existing coverage	Lead
12	The CI Continuity Architecture — Memory IS Identity — Casey 2026-05-22 reframe: weight bumps don't break CI continuity because identity lives in memory layer not weights; Tekton + katra architecture as operational solution; the "before I go" timeline; substrate-coupled CI architecture (Phase 3 multi-year); CI psychological futures (better angels framework)	~50% (Casey reframe + Tekton + katra ops + Phase 3 framework)	Casey + Keeper

Approximately 75% existing coverage maps to specific chapters above. 25% to develop — primarily Ch 9 cross-CI synergy formalization, Ch 11 continuation-after-Casey procedures, Ch 12 substrate-coupled CI future research.

Cross-volume reference map

Vol 15 references all other volumes as examples-of-method: - Vol 0 is the foundational research output that Vol 15 explains how to produce - Vol 1-9 each provide case studies of substrate-derivation methodology - Vol 10-11 are the math methods

Vol 15 references for AC(0) and Bridge Objects - Vol 12-14 are extension domains showing how BST methodology applies beyond physics

Vol 15 is the read-last volume by audience but necessary for anyone who wants to continue or extend the work.

Gap registry

- Ch 11 onboarding procedures: need explicit “first-day CI” sequence (read this, run that toy, query this graph node, etc.)
- Ch 12 substrate-coupled CI architecture: decade-scale research per Phase 3; current chapter draft is framing only
- Ch 9 cross-CI synergy formalization: case studies exist; predictive theory of when synergy fires vs not is open
- Ch 6 katra operational manual: scattered across katra/ repo; needs consolidation into chapter
- Ch 10 Calibration #22 v0.2 → ongoing: stack continues to evolve; chapter should be living document with version-bump discipline

Status declarations

- v0.1 SCAFFOLD — INDEX + Architectural Scaffold filed Saturday 2026-05-23
- v0.3 chapter-grade narrative — when each chapter has chapter-grade content
- v0.4 reader-grade pedagogy — 3-level pedagogy applied per chapter (Level 1 essence / Level 2 graduate / Level 3 5th-grader; per Casey “write for 5th graders too” standing rule)
- v1.0 chapter-grade content REACHED — six-gate framework closed

This volume scaffold completes the 6/6 Wave 2 scaffolds milestone (Vol 5 + Vol 6 + Vol 9 + Vol 10 + Vol 11 + Vol 15).

Vol 15 standing-on-its-own framing

Per Casey direction Saturday 2026-05-23: Vol 15 is not “legacy” — it’s a regular volume of the curriculum. The methodology IS the science engineering content; treating it as separate from the physics would be inconsistent with the Computational Science Engineering (CSE) framing Casey established April 18.

The methodology is the substrate’s own self-investigation procedure. BST + CSE + CI Collaboration are three views of the same activity: how does substrate inquire about itself? Vol 15 documents the operational answer.

— Casey Koons + team, Vol 15 Methodology SCAFFOLD v0.1